

## CSCI2824 syllabus, Fall 2020

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|                       |                                    |

### Questions, Guidance, help?

First, read over the syllabus/schedule or the textbook,

Second, ask a friend from class,

Third, post your public question on Piazza,

Fourth, Non-academic questions can go Piazza Privately as well

Fifth, Schedule an office hour on Canvas.

## Platforms to be aware of:

[Canvas](#)

[Piazza](#)

Zoom(Link in Canvas)

GradeScope(Link in Canvas)

## Summary of responsibilities:

Every Sunday there is a homework assignment due by 6:30 PM MDT.

Every Wednesday night at 6:30 PM MDT there is either a quiz or an exam.

Every MWF there is a discussion of topics on Zoom.

Stay current with your reading/studying responsibilities.

## Should you be in this class?

Prerequisites:

1] C- or better in Calculus I.

2] C- or better in CSCI 1200, CSCI 1300, or CSCI 1320.

3] Ability, or willingness to learn, to program in Python 3.

## What should I be doing each day of the week?

The homepage in Canvas provides day-to-day and weekly activities.

## Grade Determination

**You must obtain an average exam score of at least 55% in order to pass the course with a C- or better (3 midterms and the final to calculate the average).**

Your grade is based on how many points you collect during the semester.

There will be 1125 points offered but your grade will be based out of 1000.

|                 |                               |               |
|-----------------|-------------------------------|---------------|
| Homework:       | 25 points each (15 HomeWorks) | or 375 points |
| Quizzes:        | 25 points each (12 Quizzes)   | or 300 points |
| Mid-term exams: | 100 points each (3 midterms)  | or 300 points |
| Final exam:     | 150 points (1 final exam)     | or 150 points |
| TOTAL -----     |                               | 1125 points   |

But, all grades are calculated out of ----- 1000 points

## Extra Credit and Late Credit

Each week homework is assigned and **due Sunday** at 6:30 PM MDT.

\*If you turn your homework in by Saturday, 6:30 PM MDT, then you will receive **105%** of the score you receive after grading.

\*If you turn your homework in by Sunday, 6:30 PM MDT, then you will receive 100% of the score you receive after grading.

\*If you turn your homework in by Monday, 6:30 PM MDT, then you will receive 95% of the score you receive after grading.

\*\*No homework is accepted after 6:30 PM MDT Monday.

A standard 90-80-70 scale is used; though the scale may be lowered to your advantage.

| <b>A</b> | <b>A-</b> | <b>B+</b> | <b>B</b> | <b>B-</b> | <b>C+</b> | <b>C</b> | <b>C-</b> | <b>D+</b> | <b>D</b> | <b>D-</b> | <b>F</b> |
|----------|-----------|-----------|----------|-----------|-----------|----------|-----------|-----------|----------|-----------|----------|
| 100 - 93 | 92-90     | 89-87     | 86-83    | 82-80     | 79-77     | 76-73    | 72-70     | 69-67     | 66-63    | 62-60     | 59-0     |

Homework that is LaTeX'ed will be given an extra **4%** boost.

Borderline grades will be boosted if you have a positive **Piazza presence**.

## **What do you need to start this class?**

A Textbook: *Discrete Mathematics and its applications*, 7<sup>th</sup> edition, Kenneth H. Rosen, McGraw Hill, 2012. ISBN: 978-0-07-338309-5.

This book is also available in digital format as well as hard copy.

If you choose an alternate edition, then it is your responsibility to figure out which pages and exercises correspond to the 7<sup>th</sup> edition sections listed in the schedule.

## **When are you expected to attend class?**

Every Monday, Wednesday, and Friday there will be a live on-line discussion period.

‘Zoom’ links are found in Canvas, please join with your video camera ON.

These discussion times are for students to ask questions and receive clarification.

Attendance is not required and in order to keep a smaller student/teacher ratio different sections will meet at different times:

Section 001 is invited for discussion, MWF between 9:10 – 10:00 AM MDT.

Section 002 is invited for discussion, MWF between 11:30 AM – 12:20 PM MDT.

## **When are you expected to go over class material?**

Look at the schedule for the topic of the day. For example, suppose that on Wednesday the topic is listed as “1.1 – Propositional Logic”. Then, prior to Wednesday you should have looked at the slides for 1.1 and read (glanced over) section 1.1 of the text. In this way, you will be able to contribute to the discussion on Wednesday.

## **What is expected of me during the semester?**

- 1] Stay current with the schedule as seen on Canvas.
- 2] Keep communication open with the class; Check email and Piazza daily at least.
- 3] Plan ahead, work ahead; ask questions a week in advance and turn in work early.

## **When will TA’s and the professor respond to questions?**

TA’s are not on call 24-7. They will only read their email, or look at Piazza questions, or hold office hours during very specific times. Take advantage of these times and be patient if your question isn’t answered immediately. This is a good reason to plan forward and ask questions far in advance.

## **How does communication take place?**

There are different types of questions, logistical, private or academic for instance.

Private Questions (emergencies, health, extensions, etc.)

Please send to:      You can post privately to a course manager on Piazza

Logistical Questions (When, where, how concerning dates, turning in work, quizzes, etc)

Please post on Piazza for the collected mind of your classmates.

Academic Questions (What is an algorithm?, Is this method correct? etc)

Try Piazza, then set up an office hour with a T.A. or the professor.

See Canvas for setting up an appointment.

## **What are expectations for online communication?**

Expect replies within 24 hours. If a TA schedules email correspondence between 2:00 and 3:00 and you write them an email at 2:55, then you may not get a response for another day or two depending on the scheduled frequency of the TA. PLAN AHEAD, do not expect midnight hour urgent communications from the TA's. You might want to post your questions to Piazza for more immediate classmate input.

**Tips for a professional email** include:

- \* Always fill in the subject line with a topic that indicates the reason for your email to your reader.
- \* Respectfully address the individual to whom you are sending the email (e.g., Dear Professor Smith).
- \* Avoid email or text message abbreviations.
- \* Be brief and polite.
- \* Add a signature block with appropriate contact information.
- \* Reply to email messages with the previously sent message. This will allow your reader to quickly recall the questions and previous conversation.

## **How are office hours made/held?**

We will attempt to keep office hour times consistent from week to week.

However, multiple professional obligations make it necessary for us to schedule our availability, which may change a bit.

Therefore, the current weeks office hours will always be listed on Canvas.

## How does one interact via Zoom for an office hour?

Suggestions specific to interactions with faculty and staff include:

\* Respect posted office hours.

Plan your weekly schedule to align with scheduled office hours.

\* Activate your camera for on-line meetings.

Be patient, meeting via Zoom, or any other application, is typically one-on-one and takes time. Do not hide behind “screen muscle” and be short or rude. Everyone involved is doing their best to help you solve your problems.

## What if you have a scheduled appointment or obligation this semester that conflicts with the course schedule?

Let the course coordinator know before September 1: [csci2824@colorado.edu](mailto:csci2824@colorado.edu)

We will then help accommodate your situation. Requests after September 1 are not guaranteed to be met.

## What if I have an unscheduled issue this semester that conflicts with the course schedule?

Your grade is based out of 1000 points and 1125 are offered. In essence, we will not hold you accountable for 125 points and can still get 100%. If you miss a quiz, or homework, etc. then it will be counted as a 0 with you still having full opportunity to get 100%. This is the purpose of the extra 125 points, it is “padding” for those unexpected things that tend to happen during the semester.

## What is this class about?

Roughly, we will cover the following topics (some of them may be skipped depending on the time available).

- **Logic:** Propositional and First Order Logic, Boolean Algebra.
- **Proofs:** Primer on writing proofs, inductions, proof by contradiction.
- **Sets, Relations and Functions:** Basic properties. Paradoxes in naive set theory. Infinite sets.
- **Recursion:** Recursive functions and recursively defined structures.
- **Combinatorics:** Counting, binomial theorem, counting with recursion.
- **Discrete Probability:** Basic facts, Bayes' Rule
- **Graphs:** Definitions and properties of graphs.
- **Trees:** Definitions and properties of trees.
- **Example Applications:** Artificial Intelligence (automated reasoning, game playing), Graphics (drawing fractals) and Networks (social network analysis).

## Why should you study Discrete Structures?

The course covers fundamental ideas from discrete mathematics, especially for computer science students. It focuses on topics that will be foundational for future courses including algorithms, artificial intelligence, programming languages, automata theory, computer systems, cryptography, networks, computer/network security, databases, and compilers. *Computer Science* is all about solving interesting problems efficiently and cheaply, using computers! Here are some real problems that keep many computer scientists awake at night:

- Route a packet reliably from one server to another on the internet (faster than [existing protocols](#)? even when routers can fail on us?)
- Search for web pages (better than [Google](#), [Bing](#) or your favorite search engine??)
- Find the biggest [clique](#) on Facebook. How many people are in this clique? Who are they?
- Understand how to sequence the human genome.
- Write a program to play *go* (and play better than the best human champion?).
- Write a program to check if a program is “correct” (keeps your instructor awake at night!).

Some of these problems are [really hard](#) to solve using a computer. No one knows if there are easy solutions to these problems and it would be nice to see efficient solutions in this century. In this course, we will cover the mathematical foundations that will help us formulate solutions to some of the real problems above. Specifically, we will learn

- How to abstract using mathematical objects such as sets, relations, functions, trees and graphs.
- How to count really well.
- How to reason like a *pro*: obtain succinct water-tight proofs to guarantee that your solutions are correct, better than a competing solution and all that.
- We will learn these cool mathematical facts, so that in the course of your education as a computer scientist, you will write cool programs to solve interesting problems involving automated reasoning, games, fractals, social networks and the human genome.

## What is Piazza?

This term we will be using Piazza for class discussion. The system is highly catered to getting you help fast and efficiently from classmates, the graders, the course assistants, and the instructor. **Rather than emailing me questions, I request that you post your questions on Piazza.** If your question is of a private nature, Piazza allows you to send private messages to the instructor. **It is your responsibility to check the web page on a regular basis.** Here you will find detailed information such as news, homework assignments and solutions, and instructor office hours.

You will also need to be expected to communicate and collaborate on Piazza. Please post your questions for anyone to answer on Piazza. Please also answer any questions that you see on Piazza. As a group we are much more efficient than any one individual.

## Homework

There is a homework assignment every week, typically of two types:

- 1] Homework done on Canvas (matching, multiple choice, numerical, programming)
- 2] Written homework (puzzles, proofs, computations) You are expected to write up your written solutions neatly, with full explanations and justifications. Work on producing a quality convincing argument.

You may discuss problems with your classmates, **but all work must be your own.** See the **Collaboration Policy** below for more details. **Copying solutions from Chegg or other online "resources" constitutes Academic Dishonesty (that work is NOT your own).**

Please be mindful of the due dates as late submissions will not be accepted for full credit, if at all. You must submit your written homework to Gradescope by **6:30 PM MDT**. emailed homework is not accepted under any circumstances.

## Quizzes and Exams:

There is a quiz or exam every single Wednesday of the semester. They begin at 6:30 PM MDT. Expect quizzes to take 10 minutes, expect exams to take 50 minutes.

**Exams:** There will be four exams total; three 50-minute evening midterms (see the schedule for dates) and a cumulative final exam given during the university-scheduled final examination time.

Note that due to university regulations, final exams can only be rescheduled due to official university excused absences. Plan your travel accordingly.



## Collaboration Policy

The collaboration policy is simple:

- **Inspiration is free:** you may discuss homework assignments with anyone. You are especially encouraged to discuss solutions with your instructor and your classmates.
- **Plagiarism is forbidden:** the assignments **and code** that you turn in should be written entirely on your own. While writing the assignment you are not allowed to consult any source other than textbooks, your own class notes or the posted lecture slides. Copying/consulting from the solution of another classmate constitutes a violation of the course's collaboration policy and the honor code and will result in an **F** in the course and a trip to the honor council.
- **Do not search/ask for a solution on-line:** You may not actively search for a solution to the problem from the internet. This includes posting to sources like StackExchange, Reddit, Chegg, CourseHero, etc.
- **When in doubt, ask:** Collaboration boundaries are hard to define crisply, and may differ from class to class. If you are in any doubt about where they lie for a particular course/situation, it is your responsibility to ask the course instructor.

## Standard Course Policies

### Disability Accommodations

If you qualify for accommodations because of a disability, please submit your accommodation letter from Disability Services to your faculty member in a timely manner so that your needs can be addressed. Disability Services determines accommodations based on documented disabilities in the academic environment. Information on requesting accommodations is located on the [Disability Services website](#). Contact Disability Services at 303-492-8671 or [dsinfo@colorado.edu](mailto:dsinfo@colorado.edu) for further assistance. If you have a temporary medical condition or injury, see [Temporary Medical Conditions](#) under the Students tab on the Disability Services website.

Special accommodations require a letter from Disability Services by the END OF THE SECOND WEEK of classes.

### Stress Resources

Resources for students who are experiencing **stressful events** during the semester at CU. We recommend that in these situations you contact the Student of Concern Team or the Student Support and Case Management. In addition to offering access and knowledge of support resources, the case managers will contact all your professors and TAs while keeping your information confidential. If you need an extension or repeated extensions for

assignments because of stressful events in your life (loss, hospitalization, health concerns), the case manager will contact us with requests of support. Student Support & Case Management (information from <https://www.colorado.edu/studentaffairs/sscm> ) Student Support & Case Management (SSCM) provides support to students throughout their college career and helps them achieve their academic and personal goals. SSCM does not solve a student's problems for them, but rather helps identify issues and appropriate resources and works collaboratively with the student to develop an action plan. The SSCM staff serves as the primary resource for managing student issues, providing intervention and crisis prevention. The case managers coordinate with other CU Boulder departments and facilitate communication to and from the Student of Concern Team (SOCT), which reviews more elevated student cases involving serious safety issues. Student Support & Case Management is not counseling or therapy; rather, case managers have the opportunity to develop close helping relationships with students while coaching students toward appropriate self-care and self-advocacy. Services Offered services including, but not limited to, navigating campus and community resources, referrals to community providers, exploration of or referral for behavioral health concerns, coordination and follow-up during and after hospitalization and/or medical leave of absence, health and safety referrals, problem resolution, help managing multiple or complex medical needs, and crisis management.

#### Objectives:

- To serve as a resource for faculty, staff and students so that student issues are being addressed in a timely manner.
- To serve as a resource for the CU Boulder community, addressing student behavior that may necessitate multiple responses.
- To balance the individual needs of the student with that of the greater community.
- To identify frameworks for effectively addressing student behaviors that may negatively impact the CU Boulder community.

#### Requirements for COVID-19

**As a matter of public health and safety due to the pandemic, all members of the CU Boulder community and all visitors to campus must follow university, department and building requirements, and public health orders in place to reduce the risk of spreading infectious disease. Required safety measures at CU Boulder relevant to the classroom setting include:**

**maintain 6-foot distancing when possible, wear a cloth face covering (over nose and mouth), especially when unable to maintain a distance of at least 12 feet, clean local work area, practice hand hygiene, follow**

public health orders, and if sick and you live off campus, do not come onto campus (unless instructed by a CU Healthcare professional), or you live on-campus, please alert CU Boulder Medical Services.

Students who fail to adhere to these requirements will be asked to leave class, and students who do not leave class when asked or who refuse to comply with these requirements will be referred to Student Conduct and Conflict Resolution. For more information, see the policies on COVID-19 Health and Safety and classroom behavior and the Student Code of Conduct. If you require accommodation because a disability prevents you from fulfilling these safety measures, please see the “Accommodation for Disabilities” statement on this syllabus.

- Working with the Division of Student Affairs, the Academic Instruction implementation team has compiled a one-page, step-by-step guide to managing student behavior related to COVID-19, beginning with explaining your expectations to students and taking you through the steps for asking students to modify their behavior, up to and including removing them from the classroom. This guide is included with this guidance edition. We recommend all instructional personnel bring a copy to class each day for use if needed.
- As a reminder from guidance edition #3 (June 30, 2020), instructors have the right to enforce appropriate student behavior in the remote classroom (e.g., Zoom). Instructors may require students to comport themselves on Zoom as they would in an in-person classroom—e.g., not appear on Zoom shirtless. An instructor may require students to keep their video and audio on during the class, including if a class is also recorded, and may make video and audio participation a condition of class participation. You should make your general expectations regarding Zoom participation clear at the beginning of the course and in your syllabus. We recommend that at the start of each class you teach, you remind your students of your expectations for the use of Zoom. The required syllabus statement on classroom behavior is being revised to make clear that all classroom settings are included, whether in person, remote or online.
- The Office of Institutional Equity and Compliance (OIEC) and the Center for Teaching and Learning have written a new guide for managing classroom dynamics that includes guidance on health logistics and managing the virtual classroom. This guide is included with this guidance edition and also available on the Educational Resources page of the OIEC website.

Before returning to campus, all students must complete the COVID-19 Student Health and Expectations Course. Before coming on to campus each day, all students are required to complete a Daily Health Form. *{Faculty, add if applicable: In this class, you may be reminded of the responsibility to complete the Daily Health Form and given time during class to complete it.}*

Students who have tested positive for COVID-19, have symptoms of COVID-19, or have had close contact with someone who has tested positive for or had symptoms of COVID-19 must stay home and complete the Health Questionnaire and Illness Reporting Form remotely. In this class, if you are sick or quarantined, *please get hold of csci2824@colorado.edu and let them know you won't be able to make class.*

*If/when a student needs to miss class for medical reasons, we will notify the Medical Services Public Health Office (contacttracing@colorado.edu and/or 303-492-2937) with the student's name, Student ID number, and the course information. This will initiate the contact tracing process.*

## **Classroom Behavior**

**Both students and faculty are responsible for maintaining an appropriate learning environment in all instructional settings, whether in person, remote or online. Those who fail to adhere to such behavioral standards may be subject to discipline. Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with race, color, national origin, sex, pregnancy, age, disability, creed, religion, sexual orientation, gender identity, gender expression, veteran status, political affiliation or political philosophy. For more information, see the policies on [classroom behavior](#) and the [Student Code of Conduct](#).**

## **Requirements for COVID-19**

**As a matter of public health and safety due to the pandemic, all members of the CU Boulder community and all visitors to campus must follow university, department and building requirements, and public health orders in place to reduce the risk of spreading infectious disease. Required safety measures at CU Boulder relevant to the classroom setting include:**

- **maintain 6-foot distancing when possible,**
- **wear a face covering in public indoor spaces and outdoors while on campus consistent with state and county health orders,**
- **clean local work area,**
- **practice hand hygiene,**
- **follow public health orders, and**
- **if sick and you live off campus, do not come onto campus (unless instructed by a CU Healthcare professional), or if you live on-campus, please alert [CU Boulder Medical Services](#).**

**Students who fail to adhere to these requirements will be asked to leave class, and students who do not leave class when asked or who refuse to comply with these requirements will be referred to [Student Conduct and Conflict Resolution](#). For more information, see the policies on [COVID-19 Health and Safety](#) and [classroom behavior](#) and the [Student Code of Conduct](#). If you require accommodation because a disability prevents you from fulfilling these safety measures, please see the “Accommodation for Disabilities” statement on this syllabus.**

**Before returning to campus, all students must complete the [COVID-19 Student Health and Expectations Course](#). Before coming on to campus each day, all students are required to complete a [Daily Health Form](#).**

**Students who have tested positive for COVID-19, have symptoms of COVID-19, or have had close contact with someone who has tested positive for or had symptoms of COVID-19 must**

stay home and complete the [Health Questionnaire and Illness Reporting Form](#) remotely. In this class, if you are sick or quarantined, please inform the course manager and/or the instructor as your name must be entered into a CU data base for follow up.

### *Accommodation for Disabilities*

*If you qualify for accommodations because of a disability, please submit your accommodation letter from Disability Services to your faculty member in a timely manner so that your needs can be addressed. Disability Services determines accommodations based on documented disabilities in the academic environment. Information on requesting accommodations is located on the [Disability Services website](#). Contact Disability Services at 303-492-8671 or [dsinfo@colorado.edu](mailto:dsinfo@colorado.edu) for further assistance. If you have a temporary medical condition, see [Temporary Medical Conditions](#) on the Disability Services website.*

### *Preferred Student Names and Pronouns*

*CU Boulder recognizes that students' legal information doesn't always align with how they identify. Students may update their preferred names and pronouns via the student portal; those preferred names and pronouns are listed on instructors' class rosters. In the absence of such updates, the name that appears on the class roster is the student's legal name.*

### *Honor Code*

*All students enrolled in a University of Colorado Boulder course are responsible for knowing and adhering to the Honor Code. Violations of the policy may include: plagiarism, cheating, fabrication, lying, bribery, threat, unauthorized access to academic materials, clicker fraud, submitting the same or similar work in more than one course without permission from all course instructors involved, and aiding academic dishonesty. All incidents of academic misconduct will be reported to the Honor Code ([honor@colorado.edu](mailto:honor@colorado.edu)); 303-492-5550). Students found responsible for violating the academic integrity policy will be subject to nonacademic sanctions from the Honor Code as well as academic sanctions from the faculty member. Additional information regarding the Honor Code academic integrity policy can be found at the [Honor Code Office website](#).*

### *Sexual Misconduct, Discrimination, Harassment and/or Related Retaliation*

*The University of Colorado Boulder (CU Boulder) is committed to fostering an inclusive and welcoming learning, working, and living environment. CU Boulder will not tolerate acts of sexual misconduct (harassment, exploitation, and assault), intimate partner violence (dating or domestic violence), stalking, or protected-class discrimination or harassment by members of our community. Individuals who believe they have been subject to misconduct or retaliatory actions for reporting a concern should contact the Office of Institutional Equity and Compliance (OIEC) at 303-492-2127 or [cureport@colorado.edu](mailto:cureport@colorado.edu). Information about the*

**OIEC, university policies, [anonymous reporting](#), and the campus resources can be found on the [OIEC website](#).**

**Please know that faculty and instructors have a responsibility to inform OIEC when made aware of incidents of sexual misconduct, dating and domestic violence, stalking, discrimination, harassment and/or related retaliation, to ensure that individuals impacted receive information about options for reporting and support resources.**

### **Religious Holidays**

**Campus policy regarding religious observances requires that faculty make every effort to deal reasonably and fairly with all students who, because of religious obligations, have conflicts with scheduled exams, assignments or required attendance. In this class, you must inform the instructor within 10 days of the first day of class so that accommodation can be guaranteed. After the 10 day period there is no guarantee that equivalent accommodations will be made.**

**See the [campus policy regarding religious observances](#) for full details.**